

v1.0Deep Dive_ Tau Particle Benevolence

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Tau Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Ap-
plies to:** v1.0Deep Dive_ Tau Particle Benevolence V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscil-
lations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Tau g-2	Not measured	Belle II 2023:	a_τ	< 1.1×10^-3 — consistent with SM prediction 1.177×10^-3
Tau CP violation	Not covered	Belle II 2024: CP violation in τπππν_τ not found at 1% level. V2.0 asymmetry parameter bound added	LOW	CP violation search result constrains BSM in tau sector
Hττ spin correlations	Not covered	ATLAS/CMS 2023: Higgs CP parity confirmed even from Hττ spin correlations	MEDIUM	Higgsτtau vertex CP structure now measured — affects Higgs coupling in simulation
Tau branching fractions	PDG 2020 era	PDG 2024: BR(μνν)=17.39±0.04%, BR(ενν)=17.82±0.04%, hadronic=64.79%. 0.4% precision improvement	LOW	More precise decay channel probabilities for decay simulation

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** See table
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric ($C \cdot F \cdot S$ product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Deep Dive_ Tau Particle Benevolence V1.0 archived | V2.0 is the active specification as of June 2026